Dataset Generation Prompt Specification

Project: Agentic Web Scraper and Market Intelligence Reporter | **Project No.:** 81
Domain : Agentic AI / Python | **Academic Year:** 2025-26

1. Assignment Prompt Template Reference:

"Generate 100 mock scrape session logs across 10 industry domains with extracted row counts, duration, and report metadata"

2. Complete Production-Grade Master Prompt (Ready to Copy into ChatGPT / Claude / Gemini):

Copy the verbatim prompt below and paste directly into an LLM tool to generate 100 benchmark rows formatted exactly as dataset.csv:

```
You are an expert Data Engineer and AI Systems Benchmarking Specialist.
Generate 100 realistic mock scrape session logs for an autonomous 'Agentic Web Scraper and Market Intelligence
Reporter' (Project 81, Agentic AI / Python domain).

### TASK OBJECTIVE:
Simulate production telemetry from autonomous multi-step web scraping workflows where the agent conducts
competitive market research across 50 real industry domains in 10 sectors (IT Services, Cloud SaaS, Financial
Intelligence, Cybersecurity, E-Commerce, Healthcare, Logistics, Frontier AI, EdTech, Telecom), parses HTML
structures, validates entities with Pydantic v2 schemas, and synthesizes intelligence reports.

### EXACT SCHEMA (Strict CSV Header):
run_id,target_domain,pages_crawled,data_extracted_rows,scrape_duration_s,status,report_generated

### FIELD GENERATION RULES & REALISTIC DISTRIBUTIONS:
1. run_id: Sequential execution ID starting from RUN0501 up to RUN0600 (format: RUN followed by 4 digits).
2. target_domain: Select from real corporate domains across 10 sectors:
- IT Services: hcltech.com, tcs.com, infosys.com, wipro.com, cognizant.com, accenture.com
- Cloud SaaS: salesforce.com, servicenow.com, snowflake.com, databricks.com, mongodb.com
- Financial Intelligence: bloomberg.com, reuters.com, morningstar.com, marketwatch.com, stripe.com
- Cybersecurity: crowdstrike.com, paloaltonetworks.com, cloudflare.com, zscaler.com, fortinet.com
- E-Commerce & Retail: shopify.com, amazon.com, walmart.com, etsy.com, target.com
- Healthcare: veeva.com, iqvia.com, teladoc.com, cerner.com, pfizer.com
- Supply Chain & Logistics: maersk.com, fedex.com, flexport.com, dhl.com, chrobinson.com
- Frontier AI Research: openai.com, anthropic.com, huggingface.co, scale.com, cohere.com
- EdTech & Developer: coursera.org, udemy.com, edx.org, pluralsight.com, github.com
- Telecom & Networking: cisco.com, verizon.com, att.com, ericsson.com, nokia.com
3. pages_crawled: Integer between 10 and 500 representing distinct HTML pages discovered and crawled.
4. data_extracted_rows: Integer count of structured records extracted (200 to 10,000). Must logically correlate
with pages_crawled (avg 10–25 rows per page).
5. scrape_duration_s: Total execution time in seconds as a float rounded to 1 decimal place (30.0 to 595.0s),
correlating with pages crawled and network latency.
6. status: Operational outcome reflecting real-world web conditions:
- 'success' (~80-85%): Clean crawl, parsing, and structured extraction.
- 'partial' (~10-15%): Degraded run caused by rate-limiting (HTTP 429), partial dynamic JS rendering timeouts,
or partial robots.txt disallows. Extraction count is reduced.
- 'failed' (~3-5%): Hard failure due to IP block, CAPTCHA challenge, or HTTP 403 Forbidden. pages_crawled is
low (<25), data_extracted_rows is 0 or <100, and scrape_duration_s indicates early termination.
7. report_generated: Binary flag (1 or 0):
- 1: AI synthesized report was generated (85% of successful runs, 55% of partial runs, 0% of failed runs).
- 0: Report omitted due to low extraction volume or fatal failure.

### FEW-SHOT SEED ROWS FOR PATTERN REPLICATION:
run_id,target_domain,pages_crawled,data_extracted_rows,scrape_duration_s,status,report_generated
RUN0001,hcltech.com,432,5186,397.0,partial,1
RUN0002,tcs.com,403,9556,526.2,success,0
RUN0003,infosys.com,13,2511,343.0,success,1
RUN0004,wipro.com,433,2303,378.5,success,1
RUN0005,cognizant.com,287,9984,207.2,success,1
RUN0006,accenture.com,287,6596,348.8,success,0
RUN0007,bloomberg.com,271,5080,199.0,failed,1

### OUTPUT FORMAT CONSTRAINTS:
- Return ONLY the raw CSV block with the exact header followed by 100 rows.
- No markdown explanation, no conversational text before or after the CSV block.
- Ensure all rows strictly follow the schema with zero missing values.
```

3. Prompt Engineering Architecture & Parameters Rationale:

Parameter	Target Distribution / Constraint	Engineering & Physical Rationale
`run_id`	RUN0501 → RUN5000	Sequential primary key ensuring zero duplicates across training and test splits.
`target_domain`	50 Real Industry Domains (hcltech.com, tcs.com, bloomberg.com, etc.)	Simulates multi-tenant crawling across 10 real sectors (IT, Cloud, Finance, AI, etc.).
`pages_crawled`	10 to 500 pages per run	Reflects bounded BFS crawl depth adhering to domain-level crawl budgets.
`data_extracted_rows`	200 to 10,000 records	Strong positive correlation with pages crawled (~10–25 rows per clean page).
`scrape_duration_s`	30.0s to 595.0s (Float, 1 dec)	Reflects real wall-clock latency including polite 1.5s per-request rate delays.
`status`	success: 83% | partial: 13% | failed: 4%	Realistic anomaly modeling of HTTP 429 (rate limits), JS timeouts, and 403 blocks.
`report_generated`	Binary (1 or 0)	Conditional logic: 1 requires sufficient extracted data and non-fatal status.

4. Step-by-Step Generation & Verification Workflow:

Step 1: Copy the Master Prompt from Section 2 into your preferred LLM (ChatGPT-4o, Claude 3.5 Sonnet, or Gemini 2.5 Flash).

Step 2: If generating beyond 100 rows, request subsequent batches by specifying: *'Generate next batch starting at RUN0601 with the exact same CSV schema.'*

Step 3: Save the output directly into dataset.csv or append to existing rows.

Step 4: Run automated verification in Python: assert df['run_id'].nunique() == len(df) and assert df.isnull().sum().sum() == 0.

CURRENT SEED ROW COUNT:	500 rows (included in project folder)
SUGGESTED TARGET ROWS TO GENERATE:	4,500 additional rows (to reach 5,000 total benchmark rows)

5. Ethical Scraping & Compliance Safeguards:

LEGAL & ETHICAL MANDATE: Real-world web scraping must strictly respect target domain robots.txt directives and platform Terms of Service (ToS). Do not scrape personal data (PII) or login-protected portals. All requests must implement rate-limiting (minimum 1.5s delay) and identify the agent via a transparent User-Agent string. Compliance with India's Digital Personal Data Protection (DPDP) Act 2023 is mandatory.